

IntroToProgramming Assignment 2

GitHub - <https://github.com/JordanAherne/IntroToProgProject>

Gun range

Design Brief and Tech Spec

Intention

- Give the player an area to test out weapons

Technical Spec

- Rectangle targets
 - Change colour when hit
- Mannequin targets
 - Respawn when destroyed
- Mini game where you have to destroy as many mannequins in a time period.
 - Display score at end

Level Design Document

Game Concept

- Gun range
- Multiple targets
- Respawning mannequins mini game

Level Purpose

- Test different weapons
- Demonstrate strengths and weaknesses of each weapon

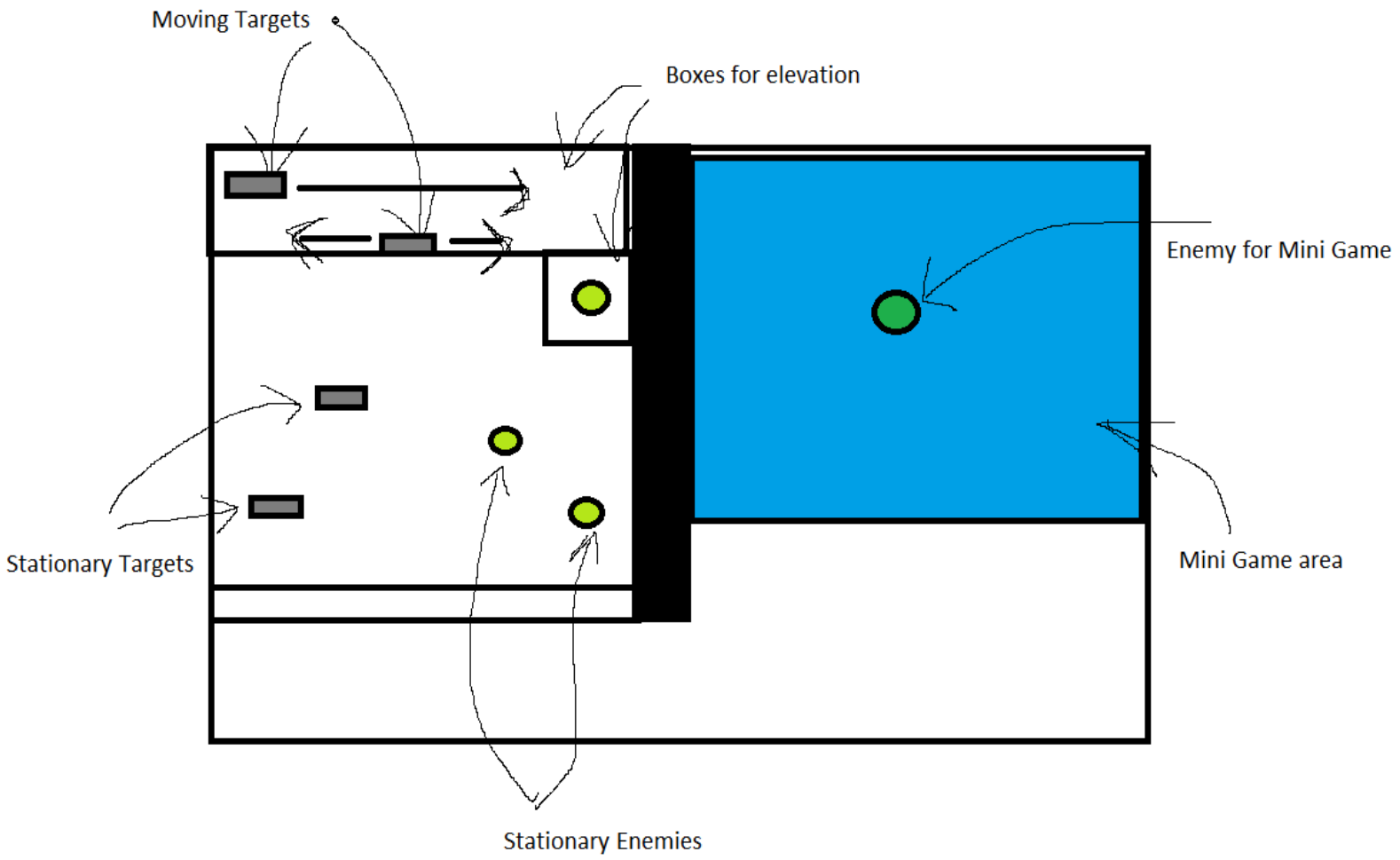
Challenges

- Shoot as many mannequins as you can in a certain time period

Elements

- Moving rectangular targets
- Stationary rectangular targets
 - Target reacts when hit
- Stationary enemy mannequins
 - mannequins have health and react when destroyed
- Respawning mannequins mini game

Level Boxing



Weapon Specs

Rifle

Auto Fire
Medium magazine
High damage
Medium fire rate
Medium range
Medium reload time

Damage	100.0
Range	10000.0
Fire Delay	0.3
Pellet Count	1
Bullet Spread	30.0
Impact Force	8000.0
Weapon Ammo	25
Max Weapon Ammo	25
Reload Time	3.0
Single Fire	☐
Auto Fire	☒

Pistol

Auto Fire
Small magazine
Low damage
Medium fire rate
Medium range
Fast reload time

Damage	25.0
Range	2000.0
Fire Delay	0.4
Pellet Count	1
Bullet Spread	25.0
Impact Force	1000.0
Weapon Ammo	15
Max Weapon Ammo	15
Reload Time	2.0
Single Fire	☐
Auto Fire	☒

Shotgun

Single Fire
Multiple pellets
Small magazine
High damage
Low fire rate
Low range
Fast reload time

Damage	15.0
Range	1500.0
Fire Delay	0.5
Pellet Count	8
Bullet Spread	50.0
Impact Force	6000.0
Weapon Ammo	4
Max Weapon Ammo	4
Reload Time	2.0
Single Fire	☒
Auto Fire	☐

Sub Machine Gun

Auto Fire
Big magazine
Medium damage
High fire rate
Low range
Fast reload time

▼ Default	
Damage	20.0
Range	3000.0
Fire Delay	0.1
Pellet Count	1
Bullet Spread	70.0
Impact Force	1500.0
Weapon Ammo	35
Max Weapon Ammo	35
Reload Time	2.0
Single Fire	☐
Auto Fire	☒

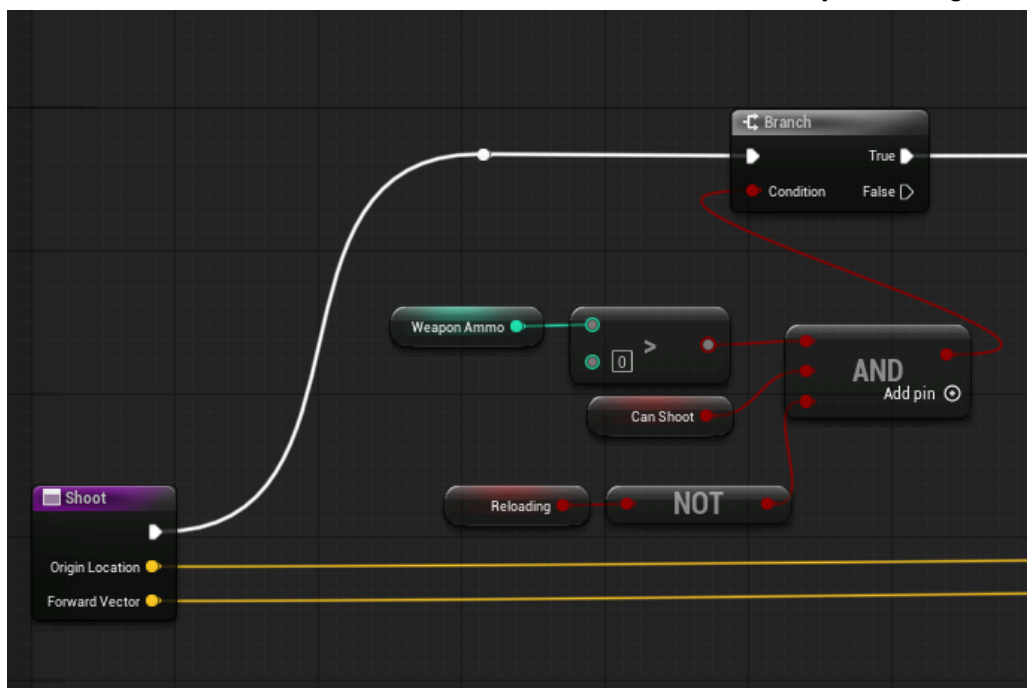
Steps Taken to Implement Spec

Weapons

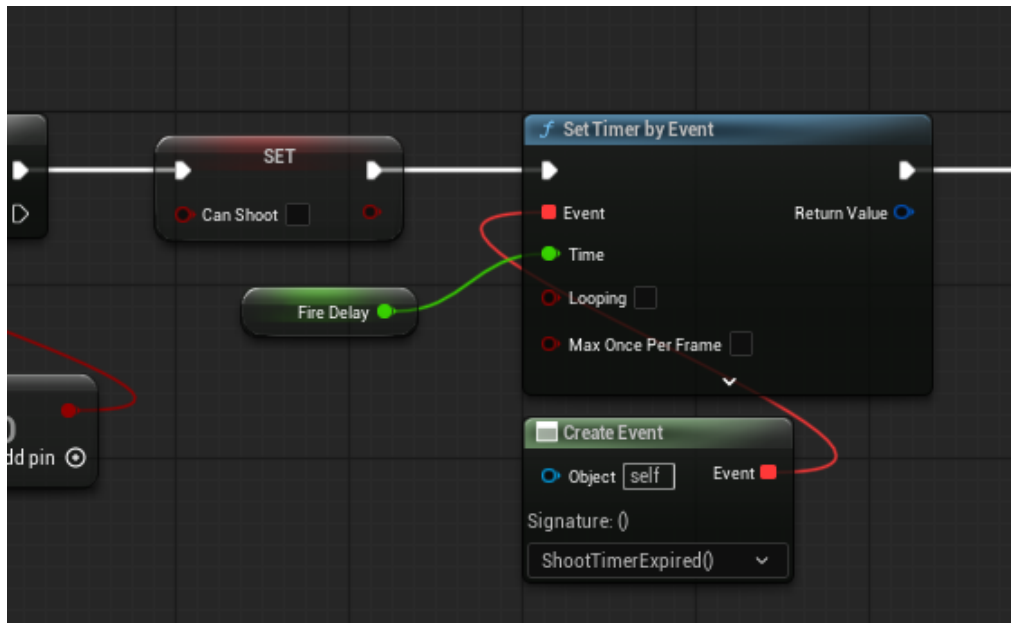
- I created BP_Weapon as my base weapon blueprint.
- Inside here I created two functions, Shoot and Reload.

Shoot

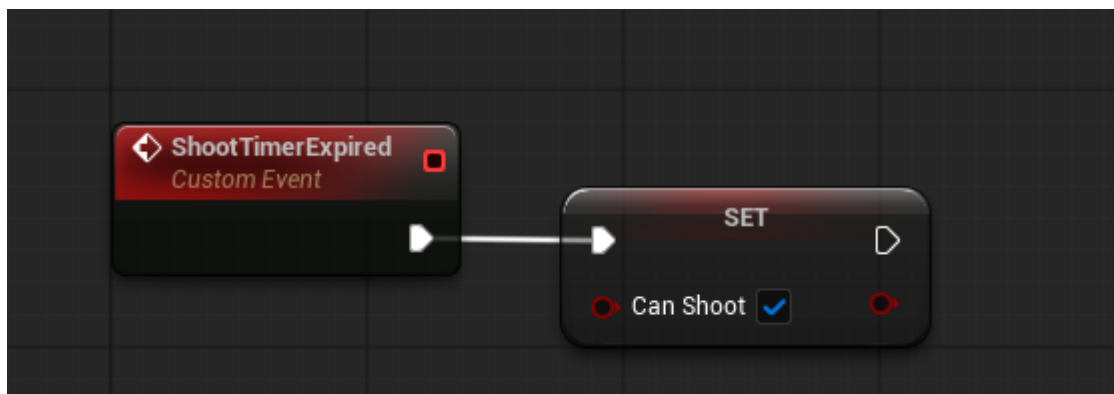
- In Shoot I pass in Origin Location and Forward Vector for the line trace
- First I check if I have Ammo, if I can shoot and if im not currently reloading



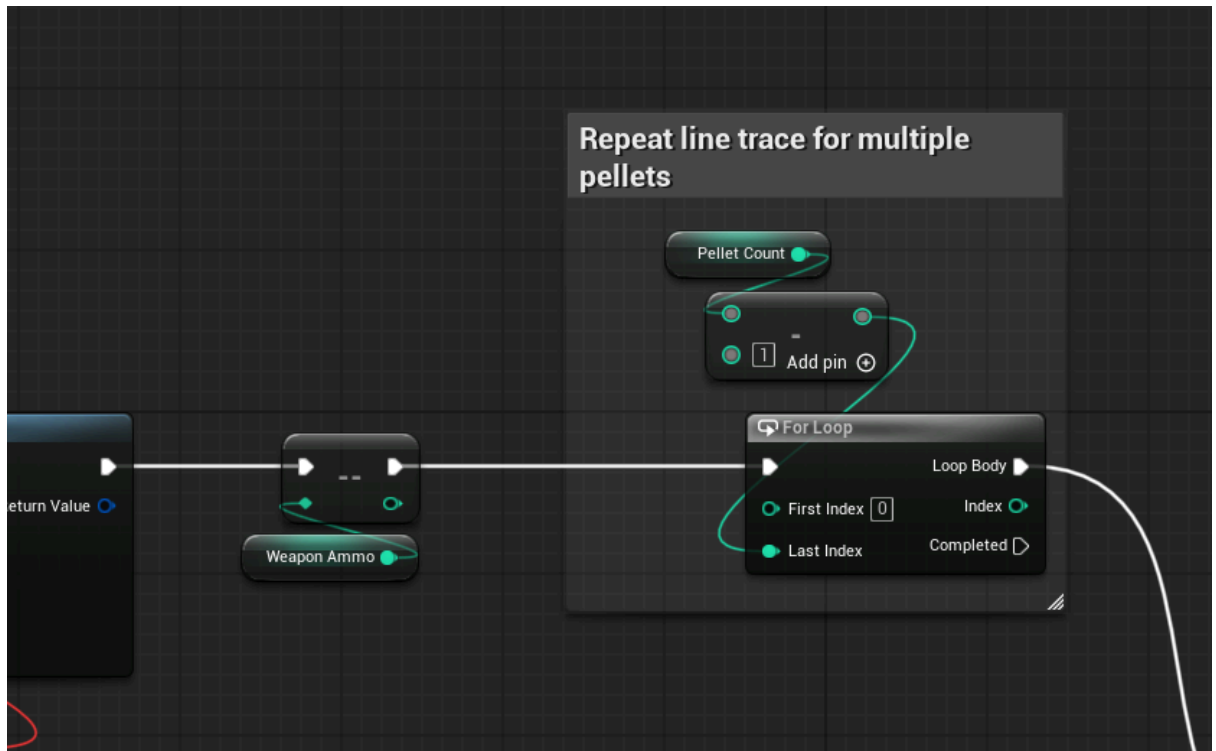
- Next I implement a delay between shots to simulate fire rate by disabling the ability to shoot and then setting a timer for the duration specified in Fire Delay.



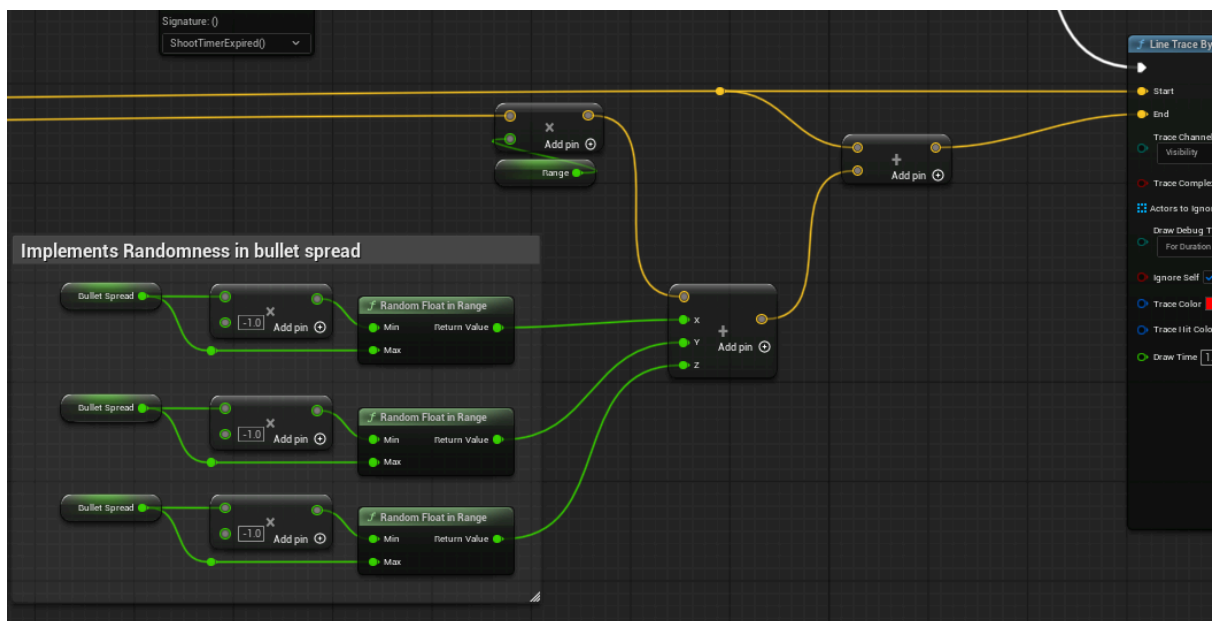
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- After the timer is up I call a custom event ShootTimerExpired to set canShoot back to true



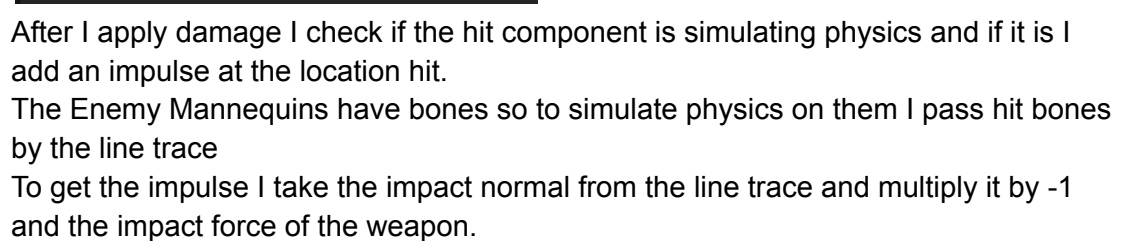
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- I then decrease WeaponAmmo and call a for loop to repeat for each pellet the weapon should fire



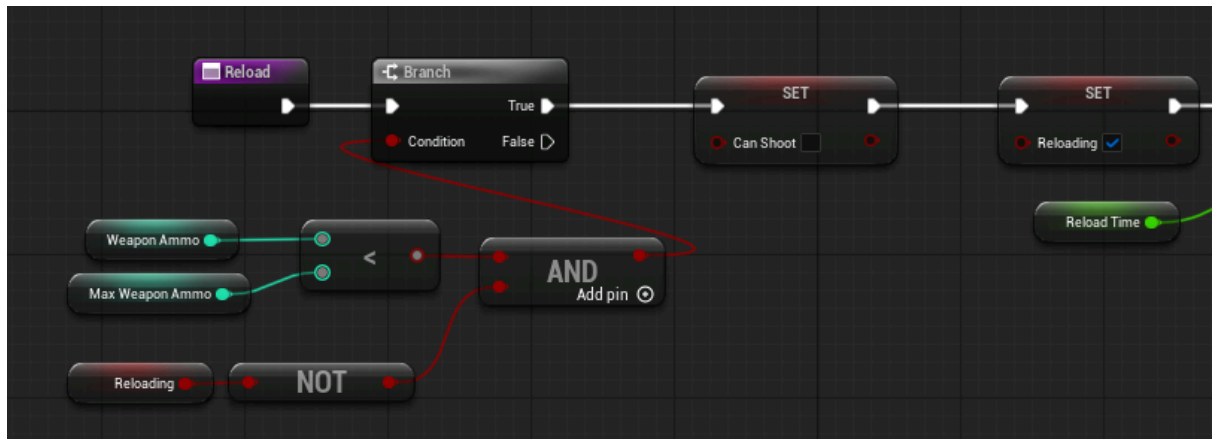
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- Inside the for loop I do a line trace for the bullets of the weapon
- For this line trace I set the start as the origin location passed into the shoot function and the end location is the forward vector multiplied by the range added to the forward vector



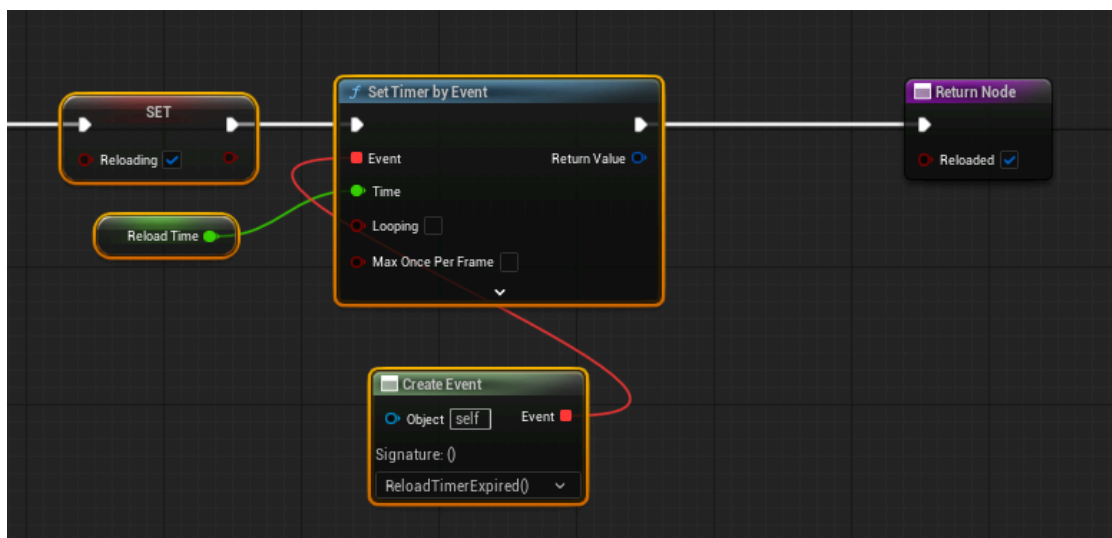
-
- Here I also implement bullet spread by adding a random value between the positive and negative value I set in bullet spread, to the forward vector.
- After the line trace I first check if the hit component is a valid object and I then use the in engine damage system to apply damage to the hit actor
- In my Enemy Mannequins I use physical materials for damage modification to have variable damage depending on where the mannequin is shot.
- So in the apply damage I set the actor hit by the line trace as the damaged actor, for the base damage I get the physical material and take the Damage Modifier and



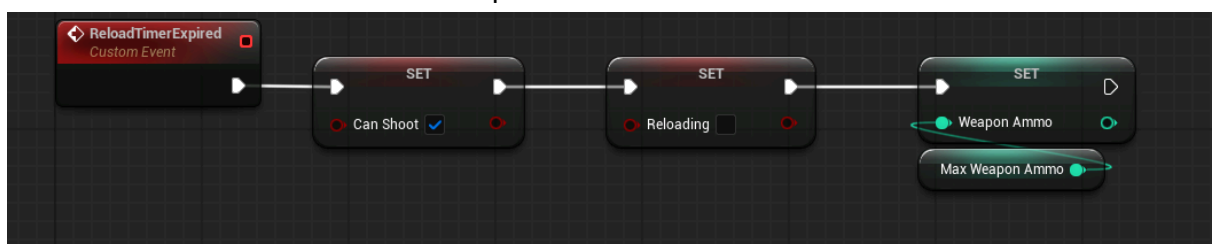
- Inside the reload function I check if I have used some ammo and if I am currently reloading.
- If true I set can shoot to false and reloading to true



- I then set a timer for the duration of reload time

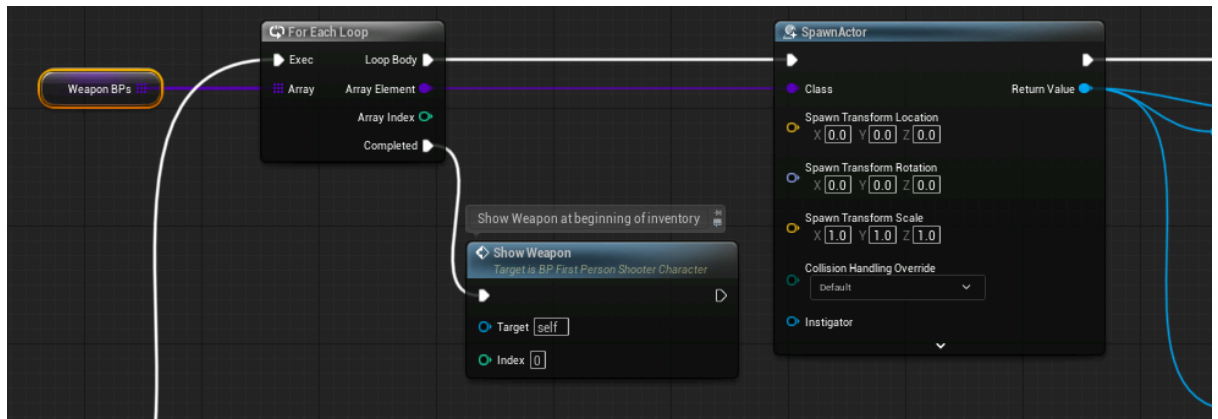


- At the end of the timer I call a custom event ReloadTimerExpired where I set can shoot back to true and reloading to false and I set the current ammo as the max ammo to simulate a reload of the weapon

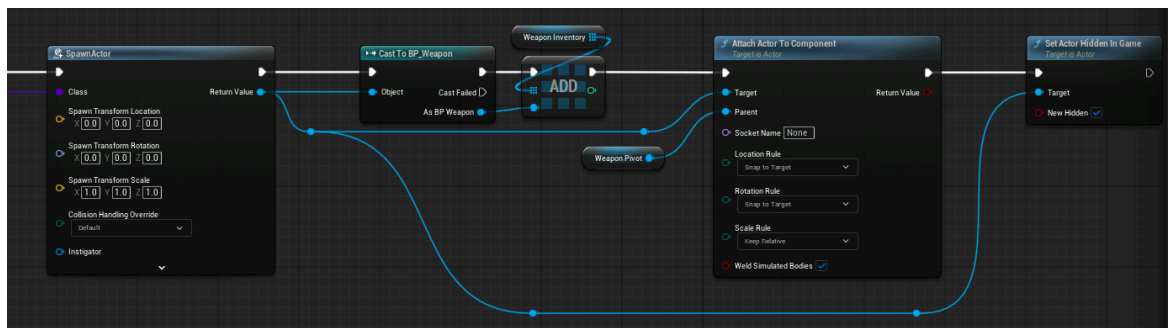


Weapons Inventory

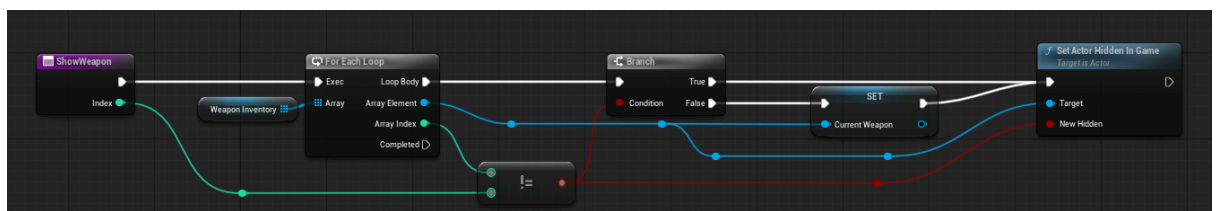
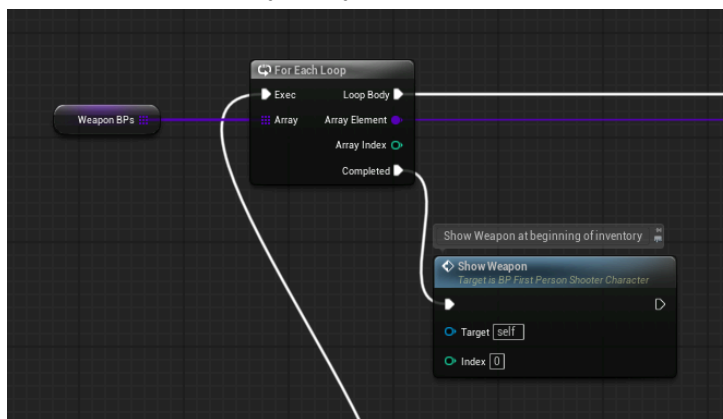
- Inside the player blueprint on event beginplay I loop through the array of actors where I imputed the Weapon Blueprints



- Inside this loop I spawn all the weapon blueprints and attach them to the actor component at the weapon pivot component
- I also add all the weapons to an array of type BP_Weapon and I set all the weapons hidden in the game.



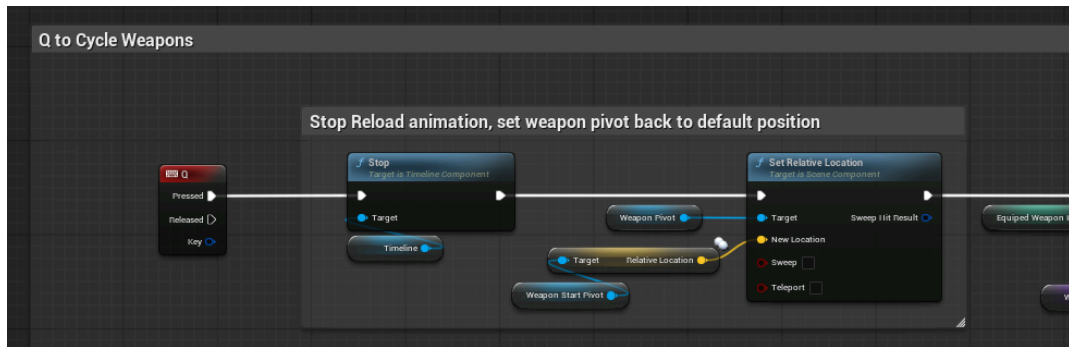
- Then at the end of the loop I call my showWeapon function to set the first element of the weapon inventory array to visible



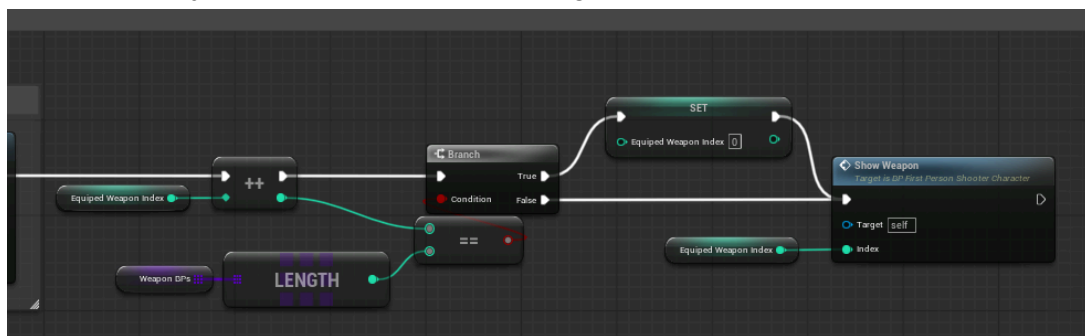
Switching Weapons

- I assigned the Q key for switching weapons

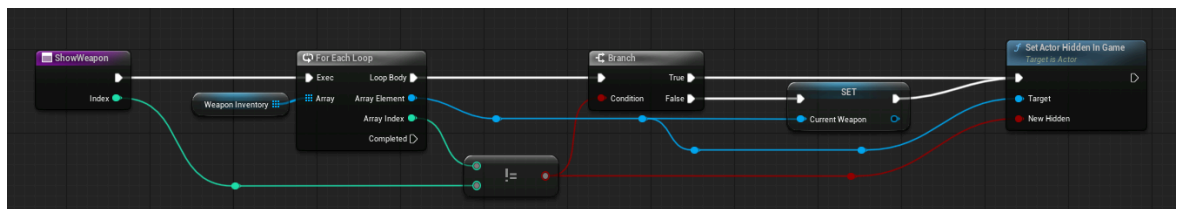
- When pressed I first interrupt and reset the reload animation



- I then increment EquippedWeaponIndex
- I check whether its larger than the length of the WeaponBPs array, if it is I set it back to 0, if its not I just call show weapon passing in the EquippedWeaponIndex



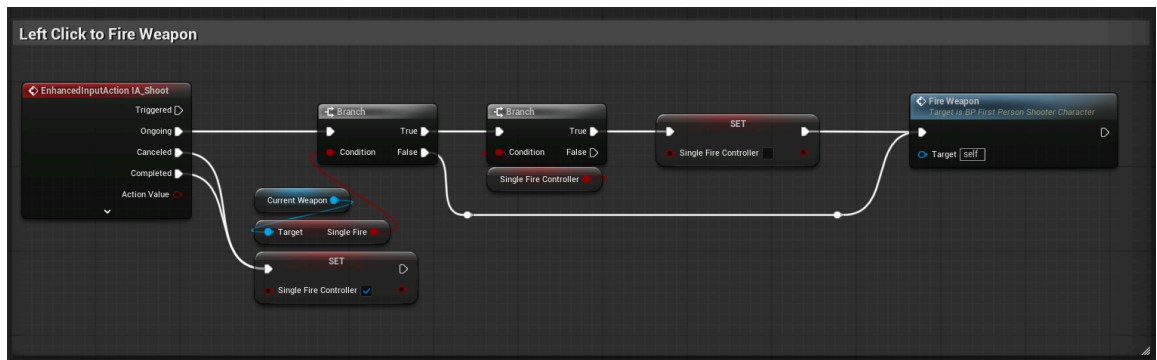
- Inside ShowWeapon I loop through the WeaponInventory array, set the current weapon variable and set the weapon at index visible.



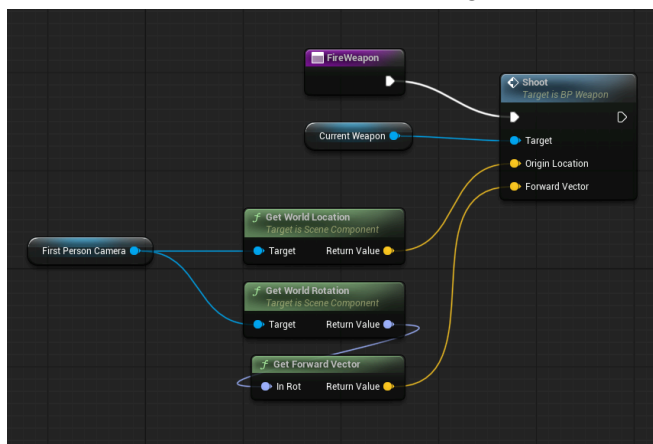
Fire Weapon

- To Fire the weapon I created a new input for the left mouse button IA_Shoot
- When called I first check whether the current weapon is single fire or not
- If it is I toggle the value of SingleFireContoller, making it so to shoot again the user must let go of left click

- If the current weapon is not single fire the FireWeapon function is repeatedly called while left click is held for auto firing

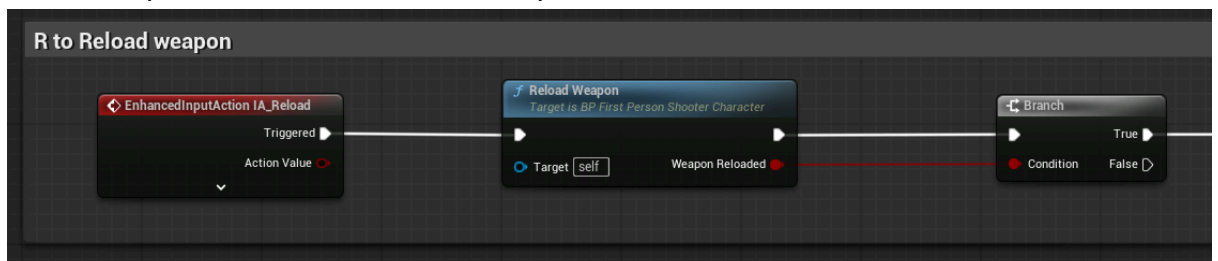


- The fire weapon function calls the shoot function on the gun passing in vectors for the first person character for drawing the line trace

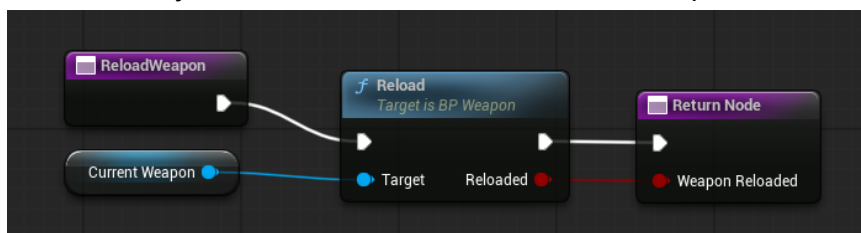


Reload Weapon

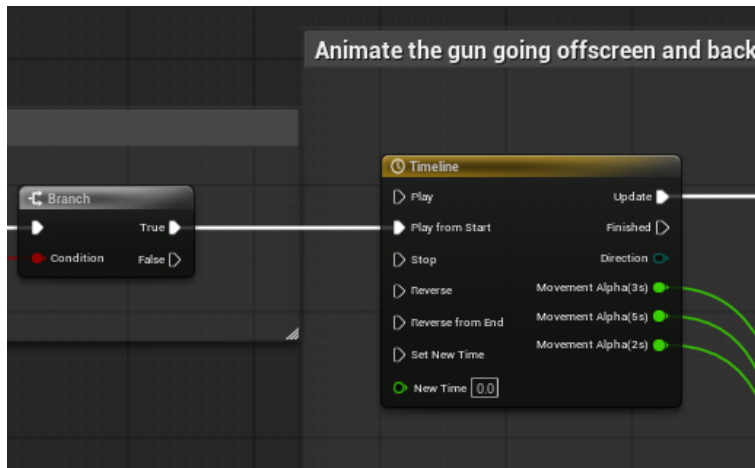
- To Reload the weapon I created a new input for the R button IA_Reload
- When R is pressed I call the Reload Weapon function



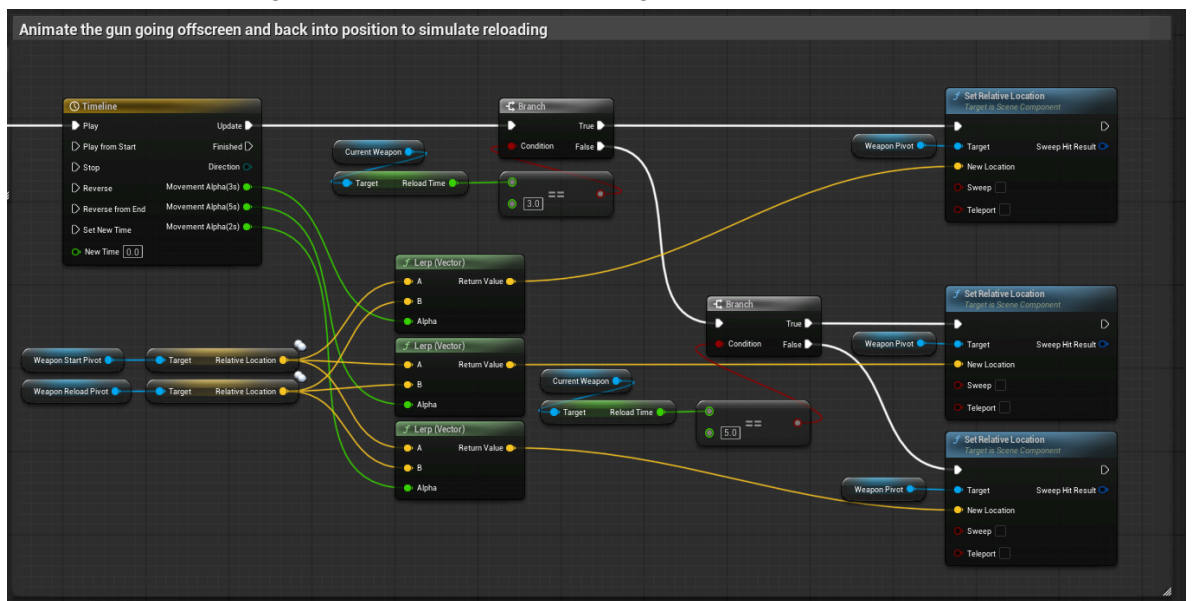
- This function just calls the reload function on the weapon



- I then check if the weapon has started reloading and if it has I create a Timeline



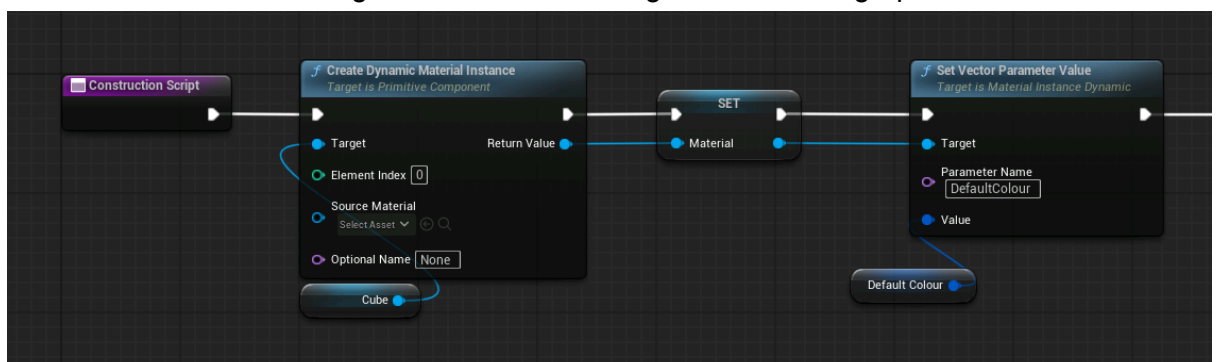
- This timeline is to animate the weapon going down off screen to simulate a reload
- I have 3 different length timelines for different length reloads



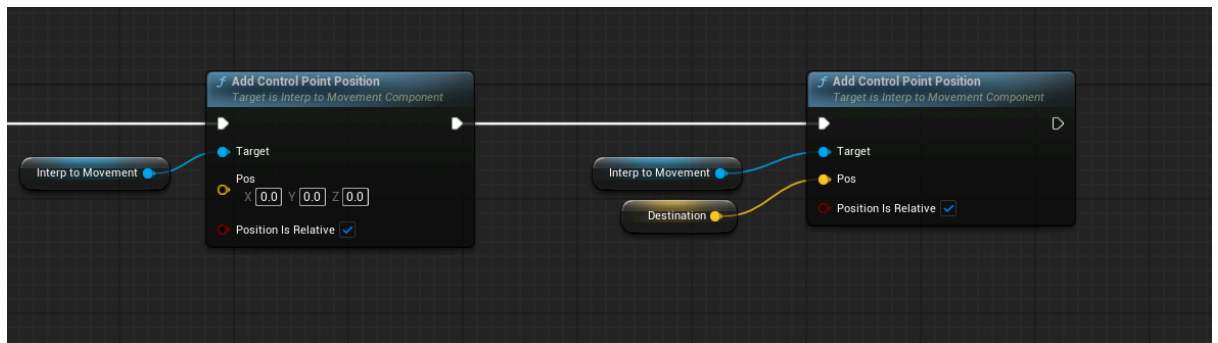
- Here I check which movement alpha to use depending on the weapon reload time.
- I then use the lerp node to interpolate between two weapon pivots I set up on the player mesh and i set the relative location of the weapon pivot

Targets

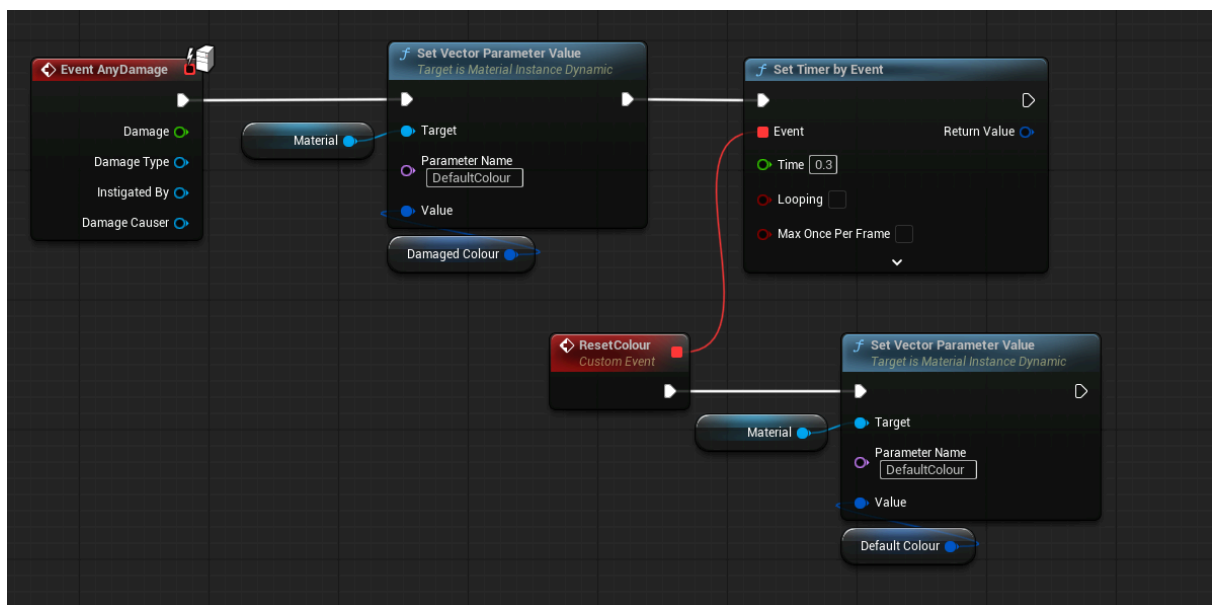
- In the construction script of the Target Blueprint I first create a dynamic material instance so that I can change the colour of the target in the event graph



- I then add two control points to the InterpToMovement component so that the target can ping pong between two points.
- I set the visibility of the destination value to public and instance editable and I show the 3D widget so i can see the destination in the map

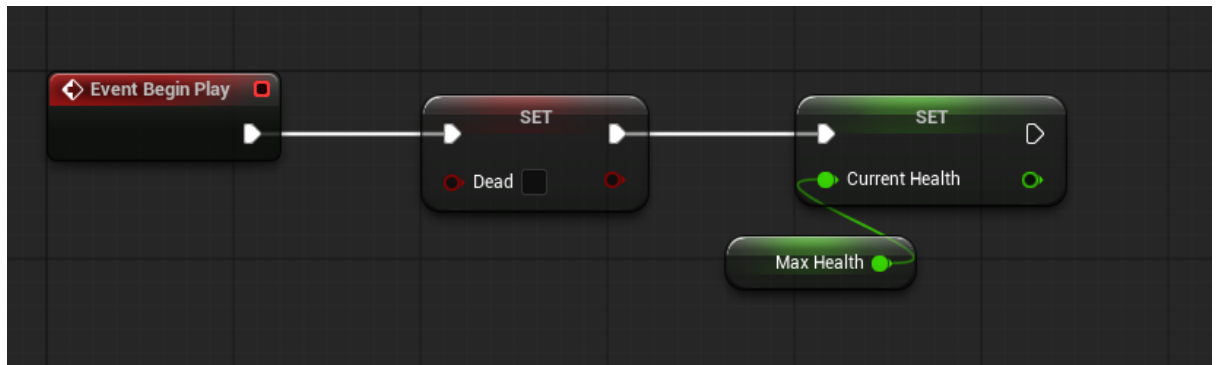


- In the event graph I call the AnyDamage event and then set the colour of the material of the target to Damaged Colour
- I do this so the target turns red when hit by the line trace of the weapon
- I then set a timer by event so that the colour of the material turns back to the default colour after 0.3 seconds.

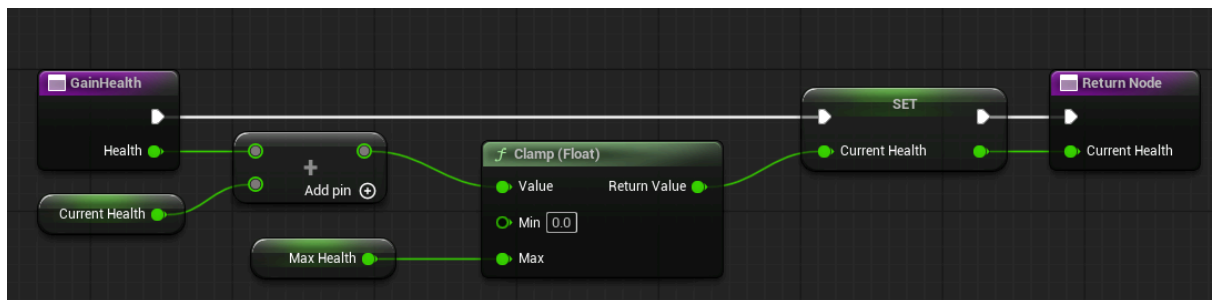


Heath Component

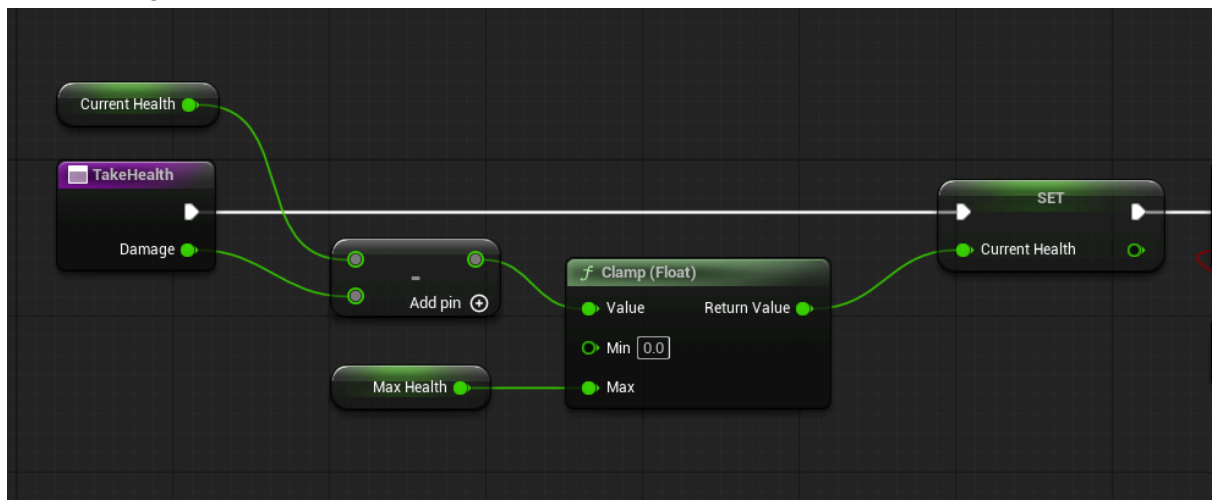
- I created a health component to add to my enemy mannequins so that damage can be dealt to them
- I created four function, GainHealth, TakeHealth, Death and HealthUpdate
- In the event graph on Event Begin Play i set the boolean Dead as false and the current Health to Max Health



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- Inside GainHealth I pass in Health and add it to Current Health, clamping the value between 0 and max health and then I set current health to the resulting value

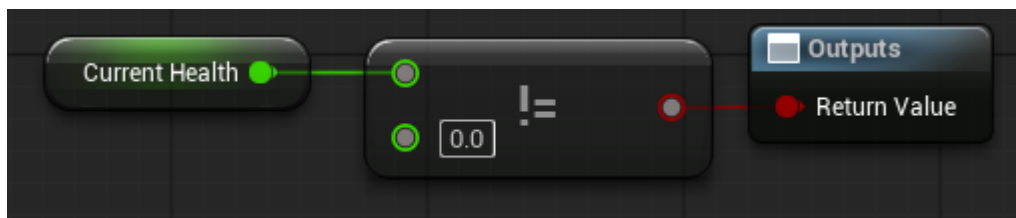
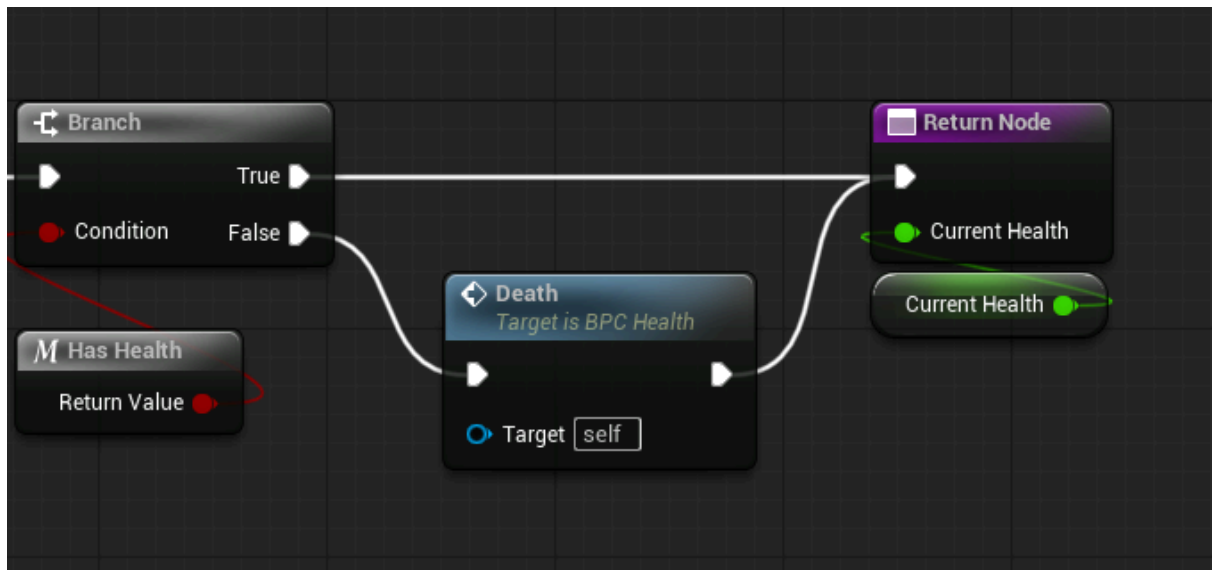


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- Inside TakeHealth I take the passed in damage value from the current health, clamping the value between 0 and max health again and then I set current health to the resulting value

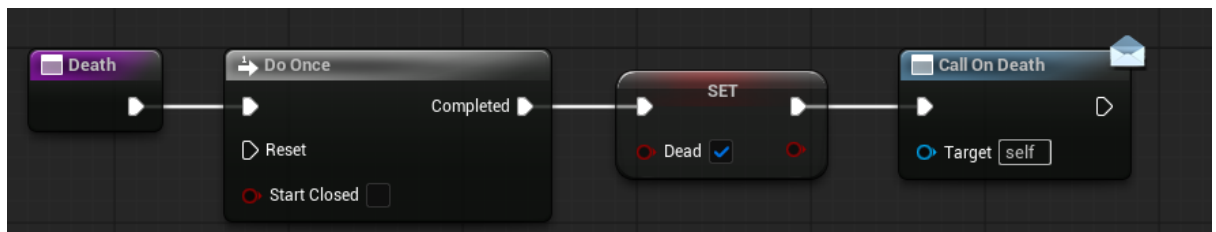


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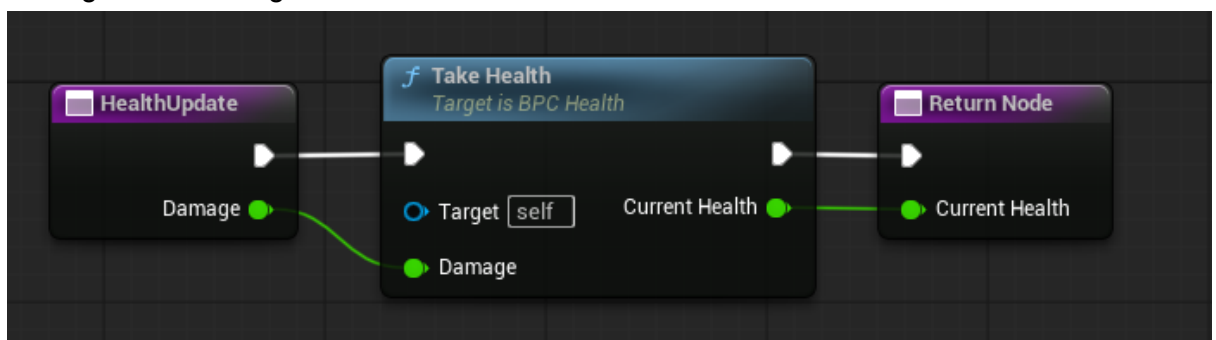
- I then use a macro to check if the actors health is greater than 0, if it is I return currentHealth, if currentHealth is 0 i call the Death function.



- Inside the Death Function I have a Do Once node to set Dead true and call the OnDeath event dispatcher.

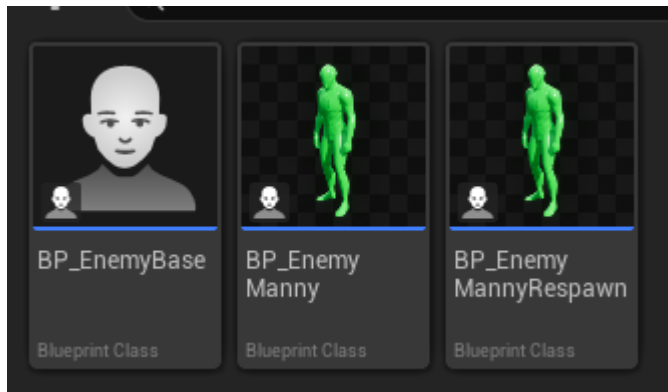


- Inside the HealthUpdate function I just call the TakeHealth function passing in Damage and returning Current Health

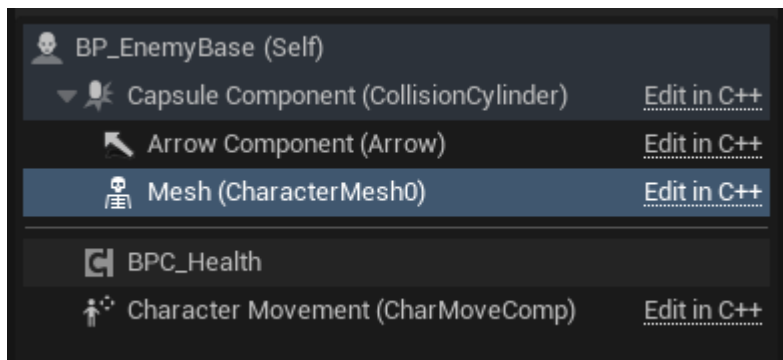


Enemy Mannequins

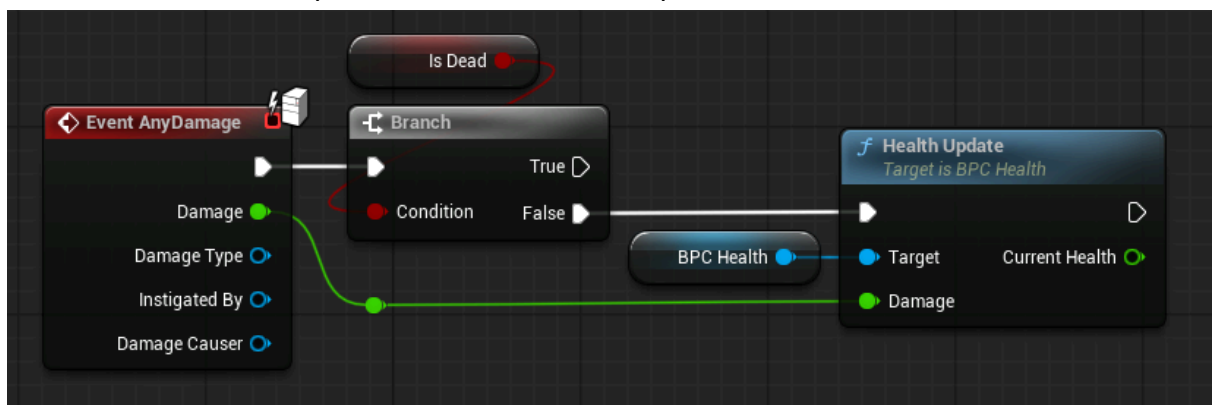
- I created an EnemyBase Blueprint to create child blueprints from for my different enemies



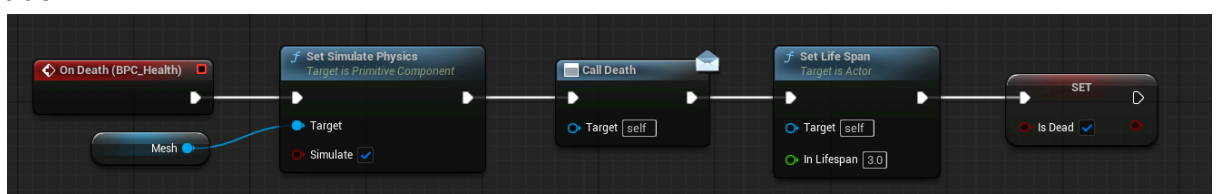
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- I first connect my BPC_Health Blueprint Component to the EnemyBase BP



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- Inside the Event Graph I add the AnyDamage event node, I check if isDead is true and if not I call Health Update from the Health component



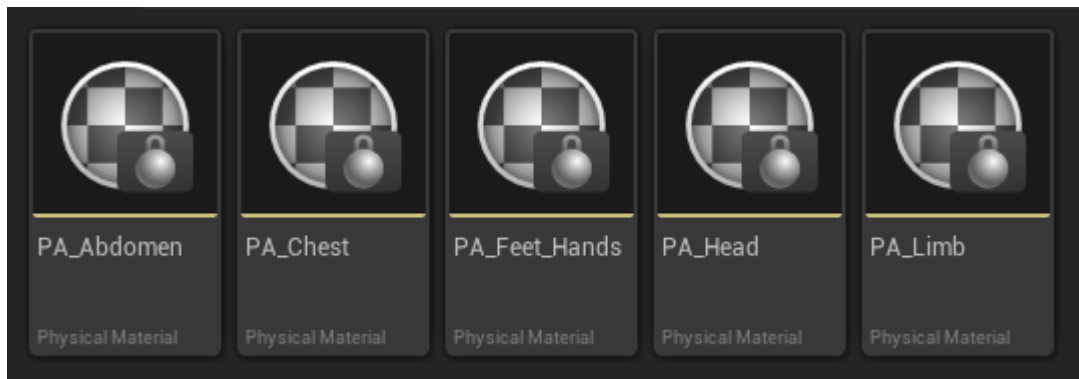
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- I also add the OnDeath event dispatcher from the health component.
- Here I set simulate physics to true on the enemy mesh, I call a death event dispatcher, I set the life span of the mannequin to 3 seconds and I set isDead to true.



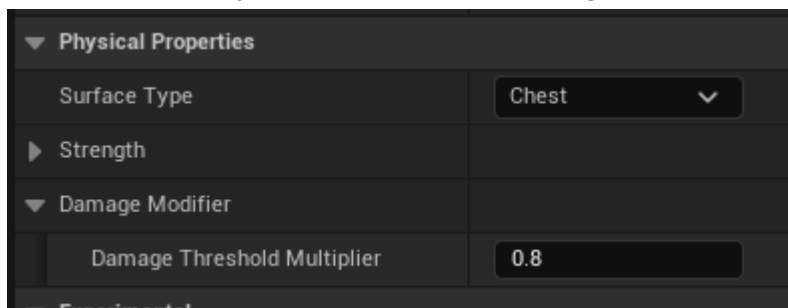
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- Note: I think technically since I have a do once in the Health component I shouldn't have to do the check for isDead on the anyDamage event but I was running into an issue where when I shot the dead mannequin it would keep calling the Death event dispatch and adding the check fixed this.

Variable Damage to mannequins

-]For the mannequin meshes added the Third Person content pack.
- To have different damage depending on where the mannequin is shot I created five physical materials



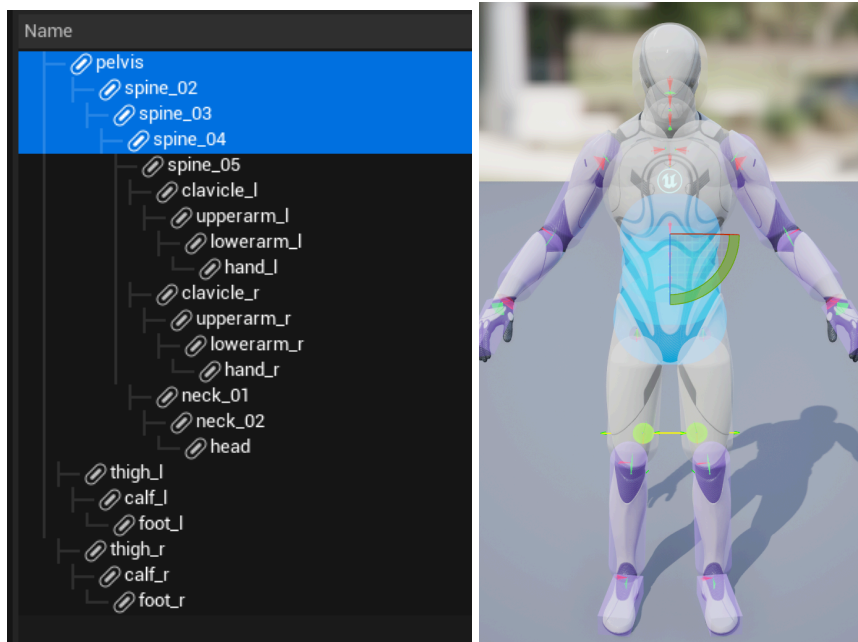
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- Inside each physical material I set the Damage Modifier depending on what body part it is.
- The head damage modifier is 1 and then each other is a lesser modifier
- This means the head will take the full damage specified by the weapon damage value and the body parts will take a percentage of that value.



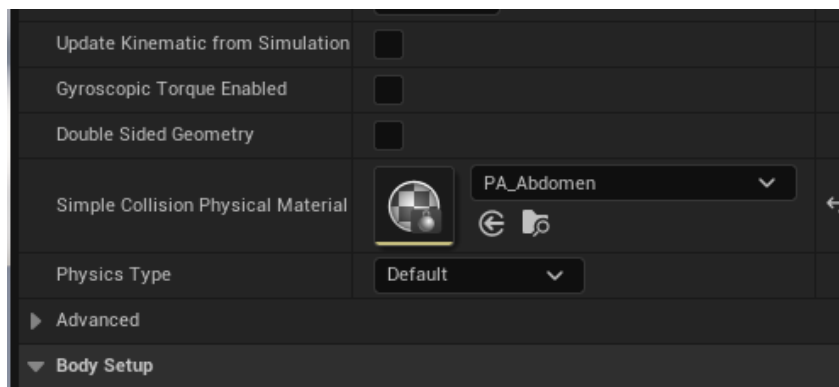
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- For example the chest takes 0.8 x the weapon damage

Editing mannequin materials

- To add my physical materials to the mannequin I edited the SKM_Manny_Simple I added by importing the content pack
- Inside SKM_Manny_Simple I clicked the PA_Mannequin button in the top right corner
- Here I can select groups of physical assets on the mannequin and add the physical materials that I created to different areas of the body



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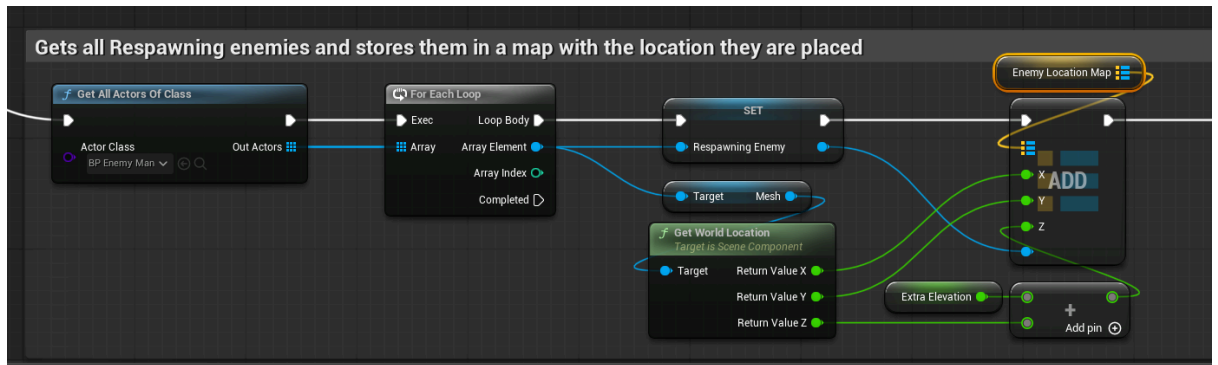
- Above I apply the Abdomen Physical material to the mannequin
- Now the mannequins take different damage depending on where they are shot.

GameMode Blueprint / Mini Game

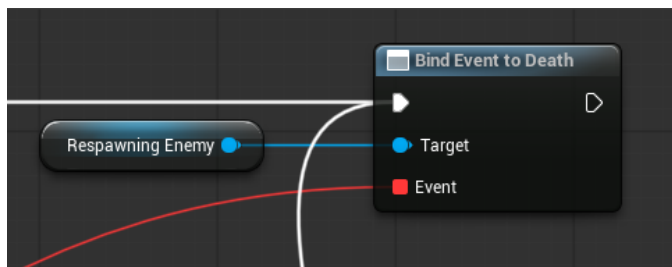
- Inside the game mode blueprint I manage two things, mannequins respawning in the target range area and the mannequin spawning for the valorant style shooting range.

Respawning Mannequins

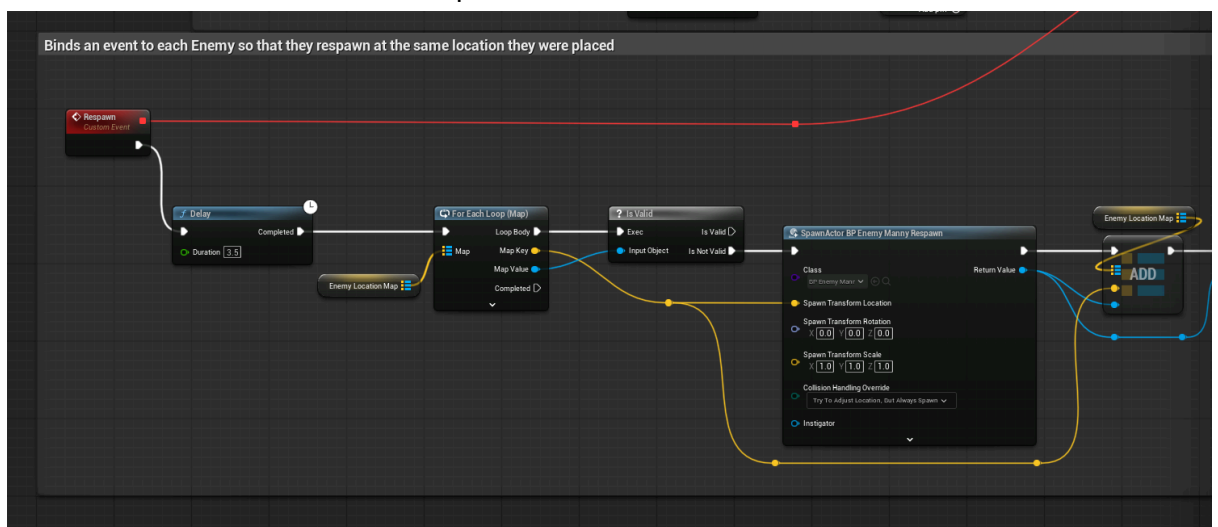
- For the respawning mannequins in the target range area I first get all actors of class EnemyMannyRespawn.
- I loop through all the results and add them to a map called Enemy Location Map.
- In this map I store the vector of their location as the key and the EnemyMannyRespawn blueprint



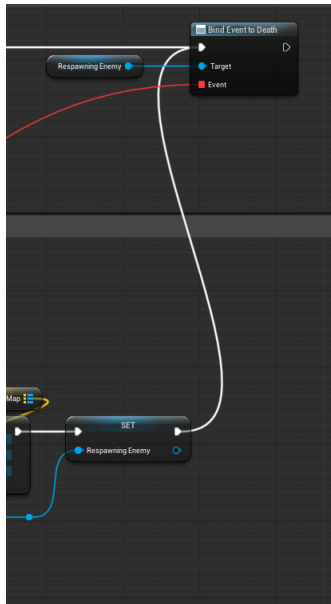
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- To each enemy I bind an event to the Death event dispatcher created in the Health Component.



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- I call the event Respawn, this event loops through the map to find which enemy was killed and then respawns an EnemyMannyRespawn actor at the location of the key in the map.
- I then add the actor back to the map.

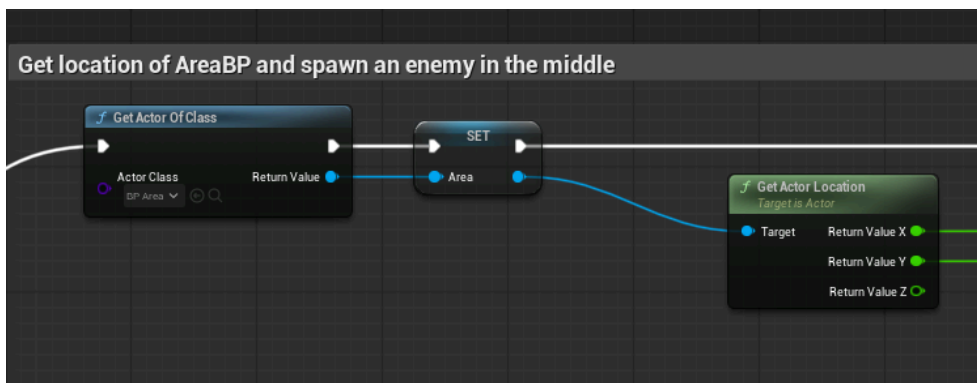


- I then loop back to the bind event to bind the same event to the newly created actor.

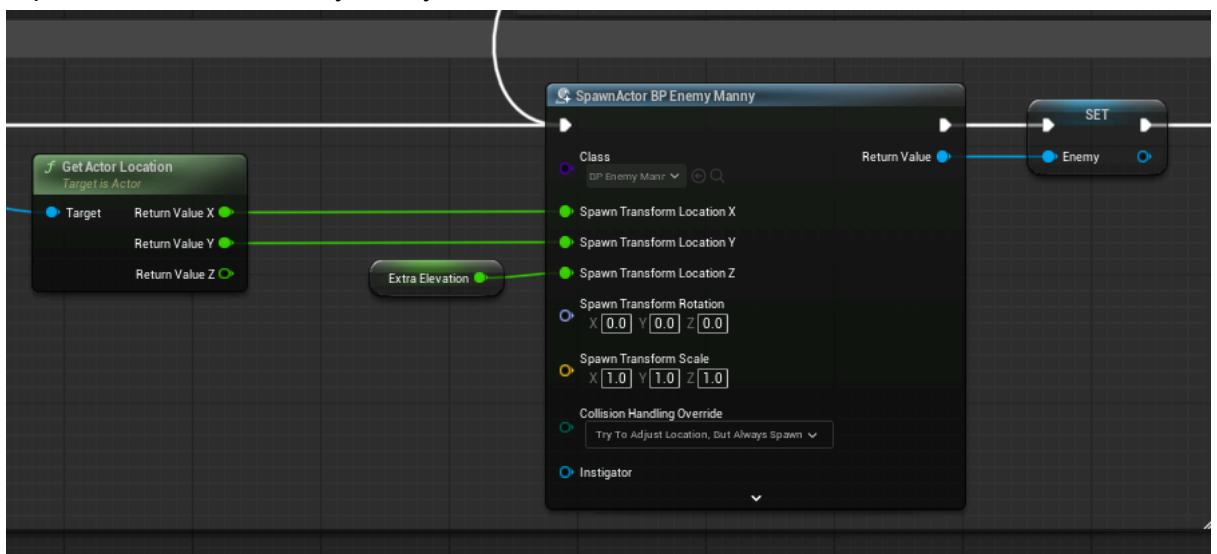


Gamemode Mannequin

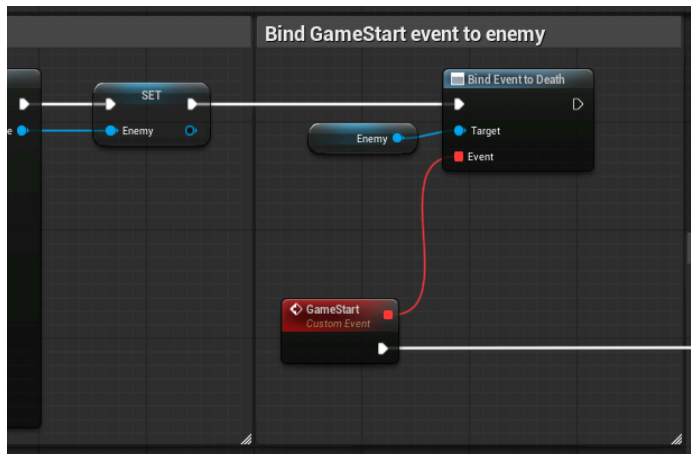
- I first get the location of BP_Area which is a plane where I want the mini game to take place



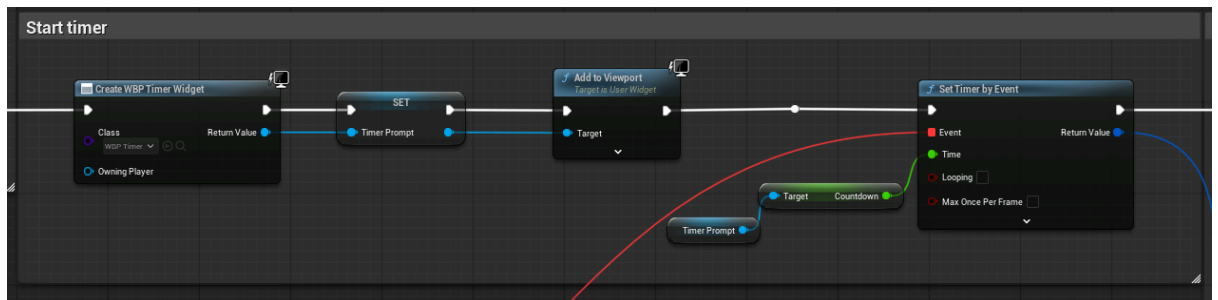
- I spawn an actor of EnemyManny in the middle of the area



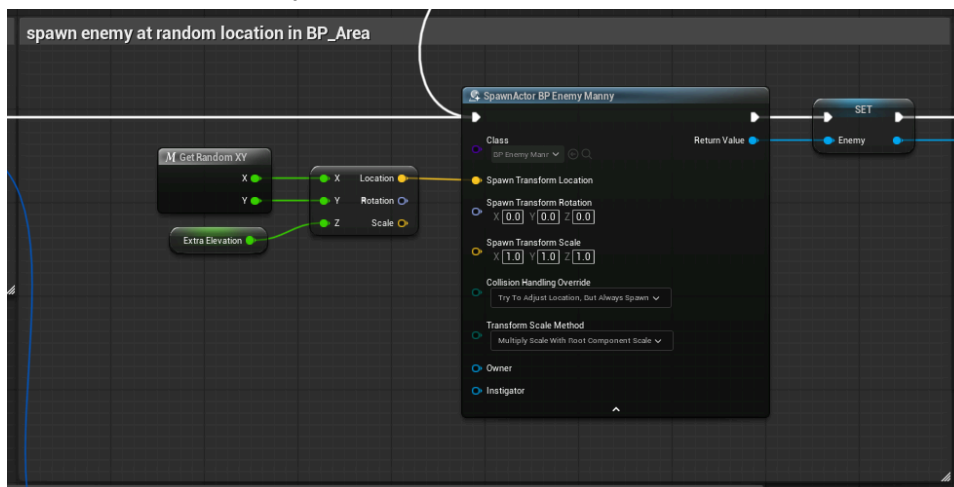
- I bind an event to this EnemyManny called GameStart



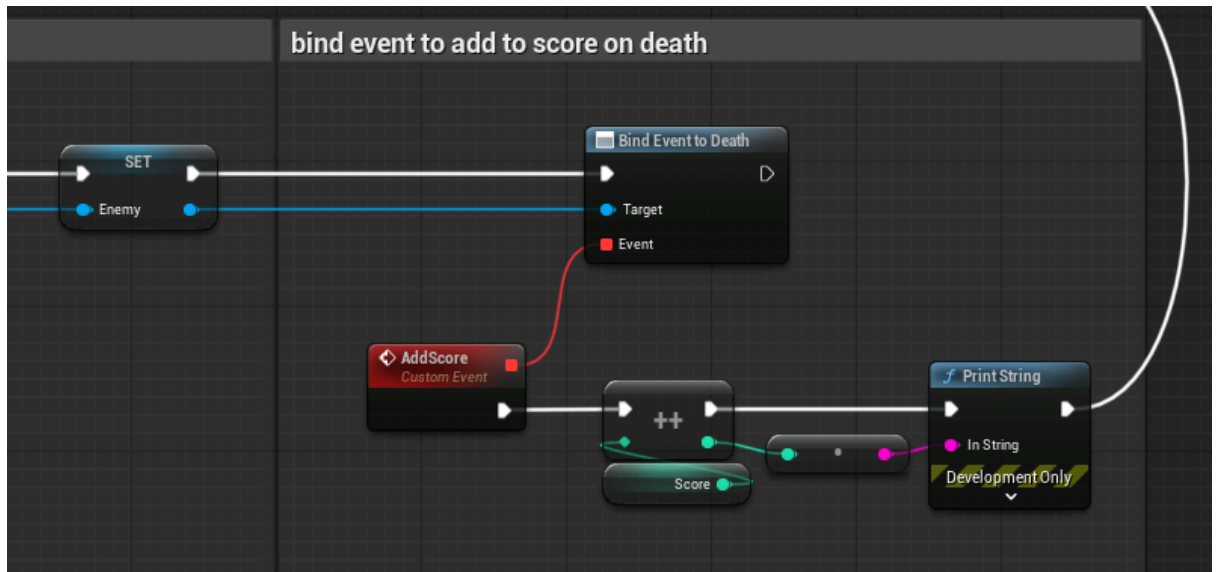
- In this event I start a timer for 30 seconds and add it to the viewport



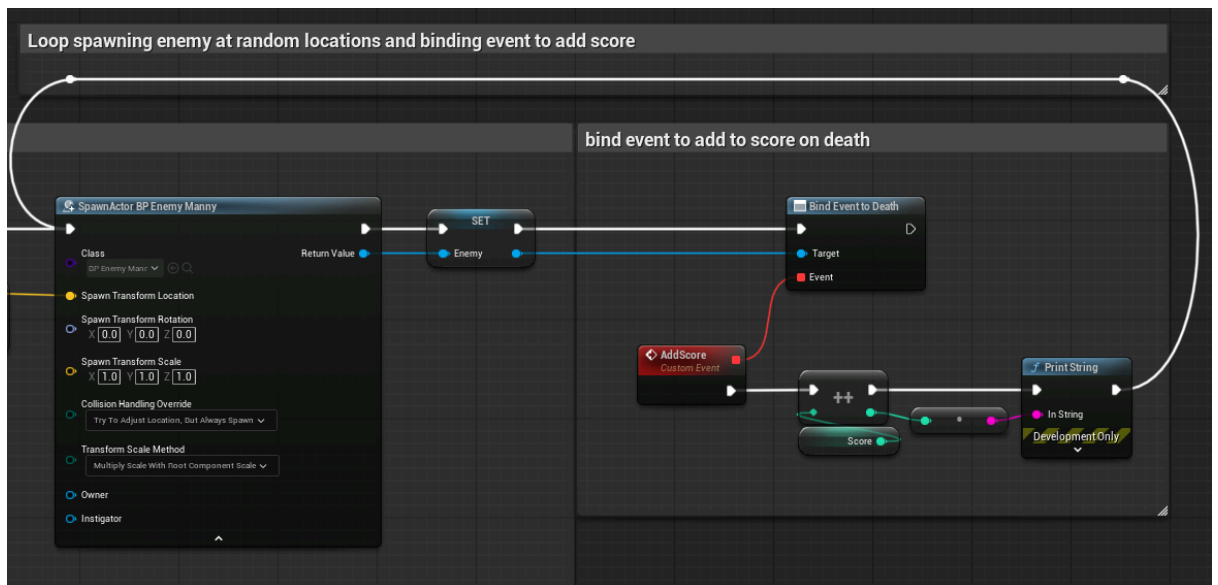
- I then spawn an enemy at a random location in BP_Area



- I bind an event to this enemy called AddScore to increase your score when killed



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- This event loops back to spawning an enemy

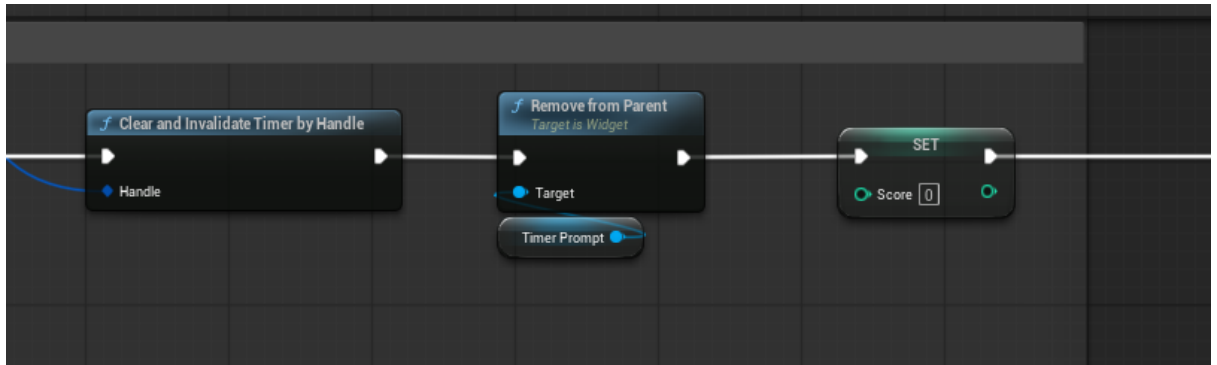


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- This loop continues until the timer is completed and the TimerEnd event is called



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- Inside this event I destroy the enemy actor, add the score visible on the timer ui and display it for 5 seconds until I clear and invalidate the timer.

- I then remove the Timer from the viewport and reset the score



- This then loops all the way back to spawning the first enemy in the middle of the area bp and binding the GameStart event to it.

